OhSINT

Open Source Intelligence

TryHackMe Room — Cyber Security Group Presentation

What we'll cover:

- What is OSINT?
- EXIF Metadata
- Digital Footprints
- OSINT Tools
- WiFi Geolocation
- Hidden Data

🕵️ WHAT IS OSINT?

Open Source Intelligence (OSINT) is the process of gathering and analysing

publicly available information to build intelligence on a target.

🔐 Security Assessments

Identify exposed info about your organisation before attackers do

🔍 Penetration Testing

Gather intel on targets during the reconnaissance phase

🧩 Digital Forensics

Trace digital footprints left behind by individuals online

🚨 Threat Intelligence

Monitor for leaks, exposed credentials, or mentions of your org

💡 *Key idea: "Any small piece of information can start a chain that reveals far more than expected."*

📁 THE CHALLENGE — ONE IMAGE FILE

The Starting Point

You are given a single file:

WindowsXP.jpg

At first glance — just a normal image. No visible clues. Nothing obvious.

The Investigation Flow:

```
Image  →  Metadata  →  Username
Username  →  Social Media  →  Location
BSSID  →  WiFi Network  →  Blog  →  Password
```

🎯 7 Questions to Answer

1. What is this user's avatar of?
2. What city is this person in?
3. What is the SSID of the WAP they connected to?
4. What is their personal email address?
5. What site did you find the email address on?
6. Where have they gone on holiday?
7. What is the person's password?

🔬 STEP 1 — EXIF METADATA WITH EXIFTOOL

What is EXIF Metadata?

Every image file stores hidden data — camera model, timestamp, GPS coordinates, author name, and more. This data is called EXIF (Exchangeable Image File Format) metadata.

ExifTool — The Tool to Use

ExifTool is a free, trusted utility for reading image metadata.

```
Install:  apt install exiftool
Run:      exiftool WindowsXP.jpg
Online:   exif.tools
```

Output from ExifTool on the challenge image:

```
GPS Latitude    : 54 deg 17' 41.27" N
GPS Longitude   : 2 deg 15' 1.33" W
Copyright       : OWoodflint
GPS Position    : 54 deg 17' 41.27" N, 2 deg 15' 1.33" W
```

🔑 **Key find: username OWoodflint**

🌐 STEP 2 — TRACING THE DIGITAL FOOTPRINT

Googling the username OWoodflint reveals THREE linked profiles:

🐦 Twitter / X

- Profile avatar → a cat 🐱
- Posted their home BSSID publicly
- BSSID: B4:5D:50:AA:86:41

🐙 GitHub

- Location listed as London 🏙️
- Email address visible in profile
- Answers Q2 (city) and Q4 (email)

📝 WordPress Blog

- Holiday destination mentioned
- Password hidden in page source
- White text on white background!

⚠️ *Lesson: A single username across platforms creates a trail. One post = multiple linked data points.*

STEP 3 — WIFI GEOLOCATION WITH WIGLE.NET

What is a BSSID?

A BSSID (Basic Service Set Identifier) is the unique MAC address of a WiFi access point — essentially your router's physical ID.

What is Wigle.net?

Wigle.net is a crowdsourced database of WiFi networks worldwide. Anyone can search a BSSID to find the network's name (SSID) and location.

How to use it:

1. Go to wigle.net → Advanced Search
2. Paste the BSSID into the search field
3. Find the network SSID from the results

3. Enter the BSSID in the BSSID/MAC field
4. Click "Advanced Search"
5. Get the network name (SSID)

The Data Chain

Twitter Post	User publicly shares their BSSID
BSSID Found	B4:5D:50:AA:86:41
Wigle.net Search	Enter BSSID in Advanced Search
SSID Result	UnileverWiFi
Location	Network pinpointed to London area

🙈 STEP 4 — FINDING HIDDEN DATA IN PAGE SOURCE

The White-on-White Trick

On the WordPress blog, a password was hidden in plain sight — the text colour was set to white (#ffffff) matching the white background, making it invisible to the naked eye.

How to find it:

Right-click the page → 'View Page Source'
Search (Ctrl+F) for keywords or scroll for suspicious hidden elements in the HTML

The HTML Source Code:

```
<p style="color:#ffffff;"
   class="has-text-color">
   pennYDr0pper.!
</p>
```

The text color matches the page background — completely invisible until you read the HTML source!

Other techniques to look for hidden content:

HTML comments · Font size 0 · Off-screen positioning · Display:none CSS · Steganography in images

OSINT TOOLS USED IN THIS ROOM

ExifTool

Reads image metadata — GPS, author, timestamps, camera info. Free & command-line.

```
exiftool filename.jpg
```

Google (Search Engine)

Search usernames, emails, or phrases. Often the fastest way to find linked profiles.

```
"0Woodflint" site:twitter.com
```

Wigle.net

Crowdsourced WiFi network database. Search by BSSID or SSID to geolocate a network.

```
Advanced Search → BSSID field
```

Twitter / X

People often overshare. Usernames, locations, network info — all found here.

```
Search @0Woodflint
```

GitHub

Developer profiles often list real location and email. Check the About section.

```
github.com/0Woodflint
```

Browser Dev Tools

View HTML source to uncover hidden text, CSS tricks, or embedded data in web pages.

```
Right-click → View Page Source
```

KEY OSINT CONCEPTS & TAKEAWAYS

Metadata Tells Stories

Images carry hidden EXIF data — GPS location, device, author. Always check metadata before anything else.

Usernames Create Trails

The same username across Twitter, GitHub, and WordPress links multiple data sources together into a full profile.

Oversharing is Dangerous

Posting a WiFi BSSID casually on Twitter allowed a stranger to identify your home location using Wigle.net.

Look Beyond the Surface

Passwords can be hidden in HTML with invisible styling. Always inspect source code — what you see isn't everything.

Information Compounds

Each clue leads to the next. A username → 3 profiles → location, email, BSSID, holiday, and password.

Legal & Ethical Use Only

OSINT on public data is legal — but using it to stalk, harass, or target individuals is not. Always act responsibly.

OhSINT

The Investigation — Solved

01

🖼️ Analyse Image

Run ExifTool on WindowsXP.jpg → Find GPS coords + username OWoodflint

02

🐦 Search Twitter

Find avatar (cat 🐱) + BSSID B4:5D:50:AA:86:41 posted publicly

03

🐙 Check GitHub

Locate city (London) + personal email address in profile

04

📡 Wigle.net

Enter BSSID → Discover WiFi SSID: UnileverWiFi

05

📝 Read Blog

Find holiday destination + inspect page source for hidden password

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